COMMUTATIVITY OF THE DUALITY DIAGRAM FOR THE TRANSPOSE CORRESPONDENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the translation of the diagram into pullbacks via the correspondence of a graph, the two explicit computations, and the identification with the graph of the morphism.

Editorial note. The label lemma-prep-dual names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-prep-dual –start 776 –end 827 –local-id 28147 returned RESULT=OK (L28147 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Correspondences of negative degree as cycle groups and composition as pullback. Same chapter, the identification of the correspondence groups from a scheme to the base with cycle groups, and the description of composition with a correspondence as pullback of cycles along it. Turns the commutativity of the diagram into an equality of two pullbacks of the fundamental cycles.
- Conversion of pushforward and pullback along a graph into the usual operations. Same chapter, the lemma describing pushforward and pullback by the correspondence of a graph in terms of the usual Gysin and pushforward maps. Identifies the two blown up expressions with the graph of the morphism.
- The Gysin pullback of the diagonal along a graph. Chow Homology, the lemma computing the Gysin pullback of the diagonal along the graph of a morphism. Supplies the last equality identifying the second route around the diagram with the graph of the morphism.

Source reference. Original lemma 0FF7: sources/stacks-chow/tags.txt, source lines 19277–19277. Prerequisite pointer invoked at proof line(s) 823; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:weil.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0FG0: sources/stacks-weil/pinned-original-source.tex, source lines 441–441. Prerequisite pointer invoked at proof line(s) 806; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FG5: sources/stacks-weil/pinned-original-source.tex, source lines 661–661. Prerequisite pointer invoked at proof line(s) 807; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : Y \to X$ be a morphism of smooth projective schemes over k. Assume X and Y equidimensional of dimensions d and e. Denote $a = [\Gamma_f] \in Corr^0(X, Y)$ and $a^t = [\Gamma_f^t] \in Corr^{d-e}(Y, X)$. Set $\eta_X = [\Gamma_{X \to X \times X}] \in Corr^0(X \times X, X)$, $\eta_Y = [\Gamma_{Y \to Y \times Y}] \in Corr^0(Y \times Y, Y)$, $[X] \in Corr^{-d}(X, \mathrm{Spec}(k))$, and $[Y] \in Corr^{-e}(Y, \mathrm{Spec}(k))$. The diagram*

$$\begin{array}{ccccc} X \otimes Y & \xrightarrow[a \otimes id]{} & Y \otimes Y & \xrightarrow[\eta_Y]{} & Y \\ {\scriptstyle id \otimes a^t}\downarrow & & & & \downarrow{\scriptstyle [Y]} \\ X \otimes X & \xrightarrow{\eta_X} & X & \xrightarrow{[X]} & \mathrm{Spec}(k) \end{array}$$

is commutative in the category of correspondences.

Proof. Recall that $\mathrm{Corr}^r(W, \mathrm{Spec}(k)) = \mathrm{CH}_{-r}(W)$ for any smooth projective scheme W over k and given $c \in \mathrm{Corr}^s(W', W)$ the composition with c agrees with pullback by c as a map $\mathrm{CH}_{-r}(W) \to \mathrm{CH}_{-r-s}(W')$ (Lemma 0FG0). Finally, we have Lemma 0FG5 which tells us how to convert this into usual pushforward and pullback of cycles. We have

$$(a \otimes \mathrm{id})^* \eta_Y^* [Y] = (a \otimes \mathrm{id})^* [\Delta_Y] = (f \times \mathrm{id})_* \Delta_Y = [\Gamma_f]$$

and the other way around we get

$$(\mathrm{id} \otimes a^t)^* \eta_X^* [X] = (\mathrm{id} \otimes a^t)^* [\Delta_X] = (\mathrm{id} \times f)^! [\Delta_X] = [\Gamma_f]$$

The last equality follows from Chow Homology, Lemma 0FF7. In other words, going either way around the diagram we obtain the element of $\mathrm{Corr}^d(X \times Y, \mathrm{Spec}(k))$ corresponding to the cycle $\Gamma_f \subset X \times Y$. □